

Date & Time Functions in Excel

Module 4: Time Creation, Conversion & Extraction Functions

In the previous modules, we covered:

- TODAY
- NOW
- DATE
- DATEVALUE
- DAY
- MONTH
- YEAR
- WEEKDAY
- WEEKNUM

Now we'll move to **Time Functions**, which are used to create, convert, extract, and analyze time values. These functions are useful in real business scenarios such as:

- Employee login and logout analysis
- Shift analysis
- Service ticket response time
- Call-center performance
- Delivery tracking
- Machine downtime analysis
- Working-hours calculation
- SLA monitoring

The functions covered in this module are:

- TIME
- TIMEVALUE
- HOUR
- MINUTE
- SECOND

10. TIME Function

The TIME function creates a valid Excel time value using separate **hour, minute, and second** values.

Purpose

It is used when hour, minute, and second values are stored separately and need to be combined into a valid Excel time.

Syntax

=TIME(hour,minute,second)

Syntax Explanation

Argument	Meaning
hour	Hour component of the time
minute	Minute component of the time
second	Second component of the time

All three arguments are required.

Basic Example

Formula:

=TIME(14,30,0)

Result:

2:30:00 PM

Here:

- 14 represents 14 hours in the 24-hour clock.
- 30 represents 30 minutes.
- 0 represents 0 seconds.

Business Example

Suppose an employee shift starts at:

Hour	Minute	Second
9	30	0

Formula:

=TIME(A2,B2,C2)

Result:

9:30:00 AM

Common Business Uses

- Creating employee shift timings
- Creating meeting schedules
- Combining separately stored time components
- Preparing machine-operation data
- Standardizing imported time records

Important Point: How Excel Stores Time

Excel stores time as a **fraction of one complete day**.

For example:

Time	Internal Value
12:00 AM	0
6:00 AM	0.25
12:00 PM	0.5
6:00 PM	0.75
12:00 AM next day	1

This is why Excel can perform arithmetic with time values.



11. TIMEVALUE Function

The TIMEVALUE function converts a **time stored as text** into a valid Excel time value.

Purpose

It is useful when time data imported from external systems is stored as text instead of an actual Excel time.

Syntax

=TIMEVALUE(time_text)

Syntax Explanation

Argument	Meaning
time_text	A text string representing a valid time

Example

Suppose A2 contains:

"10:30 PM"

Formula:

=TIMEVALUE(A2)

Excel internally returns a decimal value representing that time.

After applying time formatting, the result appears as:

10:30 PM

Business Scenario

Suppose a call-center system exports login time as text:

"08:45:00 AM"

Because it is stored as text, direct time calculations may not work correctly.

Formula:

=TIMEVALUE(A2)

This converts the text into an actual Excel time value.

Common Uses

- CSV imports
- ERP exports
- Call-center records
- Employee attendance systems
- Machine logs
- Website activity data

Difference Between TIME and TIMEVALUE

TIME	TIMEVALUE
Creates a time from separate components	Converts text into a time
Takes hour, minute, and second	Takes time text
=TIME(14,30,0)	=TIMEVALUE("2:30 PM")

Used for constructing time

Used for cleaning imported data

12. HOUR Function

The HOUR function extracts the **hour component** from a valid Excel time or date-time value.

It returns an integer from:

0 to 23

Purpose

Used to identify the hour at which an event occurred.

Syntax

=HOUR(serial_number)

Syntax Explanation

Argument	Meaning
serial_number	A valid time, date-time value, or cell reference

Example

Suppose E2 contains:

2:45:30 PM

Formula:

=HOUR(E2)

Result:

14

Excel uses the 24-hour clock for the numeric result.

Business Example Using Our Dataset

Suppose Start Time in E2 is:

10:30:00 AM

Formula:

=HOUR(E2)

Result:

10

Business Use Case: Categorize Tickets by Shift

Suppose the company follows this shift structure:

Time	Shift
Before 12 PM	Morning Shift
12 PM to before 5 PM	Afternoon Shift
5 PM onwards	Evening Shift

Formula:

=IF(HOUR(E2)<12,"Morning Shift",IF(HOUR(E2)<17,"Afternoon Shift","Evening Shift"))

This combines HOUR with IF to solve a realistic business problem.



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Common Uses

- Shift analysis
- Peak-hour analysis
- Customer-support volume analysis
- Website traffic analysis
- Employee login analysis
- Delivery-time analysis

13. MINUTE Function

The MINUTE function extracts the **minute component** from a valid Excel time.

It returns a number between:

0 and 59

Syntax

=MINUTE(serial_number)

Syntax Explanation

Argument	Meaning
serial_number	A valid Excel time or date-time value

Example

Suppose E2 contains:

10:45:30 AM

Formula:

=MINUTE(E2)

Result:

45

Business Uses

The MINUTE function can be useful for:

- Call-duration analysis
- Response-time tracking
- Machine processing times
- Employee attendance
- SLA monitoring
- Scheduling analysis

14. SECOND Function

The SECOND function extracts the **seconds component** from a valid Excel time.

It returns a number between:

0 and 59

Syntax

=SECOND(serial_number)



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Syntax Explanation

Argument	Meaning
serial_number	A valid Excel time or date-time value

Example

Suppose E2 contains:

10:45:30 AM

Formula:

=SECOND(E2)

Result:

30

Business Uses

The SECOND function is especially relevant when working with high-precision operational data such as:

- Call-center records
- Manufacturing cycle times
- Server logs
- Website events
- Financial transactions
- System response times

For many general Data Analyst roles, SECOND is less frequently used than HOUR or MINUTE, but students should still understand it.

Combining HOUR, MINUTE and SECOND

Suppose E2 contains:

10:45:30 AM

Then:

Formula	Result
=HOUR(E2)	10
=MINUTE(E2)	45
=SECOND(E2)	30

These functions only **extract components**. They don't change the original time value.

Practical Business Scenario: Calculate Ticket Completion Duration

Your dataset contains:

- Start Date in D2
- Start Time in E2
- End Date in F2
- End Time in G2

To calculate the exact total completion duration, combine the dates and times:

$$=(F2+G2)-(D2+E2)$$

This calculates:

End Date & Time – Start Date & Time

For example:

Start: 10-Jul-2026 10:00 AM

End: 11-Jul-2026 02:30 PM

The actual duration is:

1 day, 4 hours and 30 minutes

Important Formatting Point

For durations longer than 24 hours, format the result using:

[h]:mm:ss

Do **not** use only:

hh:mm:ss

because hh:mm:ss resets after every 24 hours, while [h]:mm:ss displays the total accumulated hours.

For example, a duration of **28 hours 30 minutes**:

Format	Display
hh:mm:ss	04:30:00
[h]:mm:ss	28:30:00

This distinction is extremely important in practical Excel work.

Real Business Scenario: Identify Late-Starting Tickets

Suppose company policy says tickets should start before **10:00 AM**.

Formula:

$$=IF(E2>TIME(10,0,0),"Late Start","On Time")$$

Logic

- TIME(10,0,0) creates 10:00 AM.
- Excel compares the Start Time in E2 against 10:00 AM.
- If later → Late Start
- Otherwise → On Time

This is a good example of combining TIME with IF.

Real Business Scenario: Categorize Start Time

Suppose you want:

- Before 12 PM → Morning
- 12 PM to before 5 PM → Afternoon
- 5 PM onwards → Evening

Formula:

$$=IF(E2<TIME(12,0,0),"Morning",IF(E2<TIME(17,0,0),"Afternoon","Evening"))$$

This approach is better than extracting the hour first when you're directly comparing actual time values.

Module 4 Summary



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Function	Purpose	Syntax
TIME	Creates time from separate components	=TIME(hour,minute,second)
TIMEVALUE	Converts text into a valid Excel time	=TIMEVALUE(time_text)
HOUR	Extracts the hour	=HOUR(serial_number)
MINUTE	Extracts the minute	=MINUTE(serial_number)
SECOND	Extracts the second	=SECOND(serial_number)

Core formulas from this section

=TIME(14,30,0)
 =TIMEVALUE("2:30 PM")
 =HOUR(E2)
 =MINUTE(E2)
 =SECOND(E2)

Business formulas

=IF(E2>TIME(10,0,0),"Late Start","On Time")
 =IF(E2<TIME(12,0,0),"Morning",IF(E2<TIME(17,0,0),"Afternoon","Evening"))
 =(F2+G2)-(D2+E2)

Module 5: Date Difference & Duration Functions

In the previous modules, we covered current dates, date creation, date extraction, week analysis, and time functions. Now we move to **date difference and duration calculations**, which are particularly important for Data Analysts.

These functions help answer practical business questions such as:

- How many days did a ticket take to close?
- How long has an employee worked for the company?
- How many complete years have passed between two dates?
- How many months remain until a contract expires?
- Was a service ticket completed within its deadline?
- What is the age of a customer or employee?
- How many working days were required to complete a project?

The main functions covered in this module are:

- DAYS
- DATEDIF

15. DAYS Function

The DAYS function calculates the **number of calendar days between two dates**. It subtracts the start date from the end date.

Purpose

Used to calculate duration in days between two dates.

Syntax

`=DAYS(end_date,start_date)`

Syntax Explanation

Argument	Meaning
<code>end_date</code>	The ending or completion date
<code>start_date</code>	The beginning or starting date

Important

The order of arguments matters:

End Date first → Start Date second

This is a common interview trap.

Basic Example

Suppose:

Start Date = 01-Jul-2026

End Date = 10-Jul-2026

Formula:

`=DAYS(DATE(2026,7,10),DATE(2026,7,1))`

Result:

9

There are 9 days between 1 July and 10 July.

Example Using Our Business Dataset

In our dataset:

- D2 = Start Date
- F2 = End Date

Formula:

`=DAYS(F2,D2)`

Suppose:

Start Date = 05-Jul-2026

End Date = 12-Jul-2026

Result:

7

The ticket took **7 calendar days** from its start date to its end date.

Alternative Method: Direct Date Subtraction

You can also calculate the difference using:

`=F2-D2`

This often produces the same result as:



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=DAYS(F2,D2)

Difference

Method	Example	Purpose
Direct subtraction	=F2-D2	Simple date difference
DAYS	=DAYS(F2,D2)	Explicit and readable date-difference function

For an interview, students should know both.

Business Use Case: Ticket Resolution Time

Formula:

=DAYS(F2,D2)

Possible result:

5

Interpretation:

The ticket took 5 calendar days to resolve.

Business Use Case: Check Whether Ticket Was Completed Before Due Date

Our dataset contains:

- F2 = End Date
- H2 = Due Date

Formula:

=IF(F2<=H2,"Completed On Time","Delayed")

This doesn't require DAYS, but we can calculate the exact delay:

=DAYS(F2,H2)

However, this can return a negative number if the ticket was completed early.

A cleaner business formula for **delay days only** is:

=MAX(0,DAYS(F2,H2))

Result Logic

End Date	Due Date	Result
15-Jul-2026	10-Jul-2026	5
08-Jul-2026	10-Jul-2026	0

This prevents early completions from showing negative delay days.

Important Points About DAYS

- Counts **calendar days**, including weekends.
- Does not exclude holidays.
- Requires end date first and start date second.
- Can return a negative result if the end date is earlier than the start date.
- For working days, use NETWORKDAYS instead.



16. DATEDIF Function

The DATEDIF function calculates the difference between two dates in:

- Complete years
- Complete months
- Days
- Remaining months
- Remaining days

It is particularly useful for:

- Employee tenure
- Age calculations
- Contract duration
- Subscription duration
- Customer relationship duration

Important Interview Point

DATEDIF is an unusual Excel function because it may **not appear in Excel's function suggestions or autocomplete**, but it still works in supported Excel versions.

Students often think the function is invalid because Excel doesn't always suggest it automatically.

Syntax

`=DATEDIF(start_date,end_date,unit)`

Syntax Explanation

Argument	Meaning
start_date	Beginning date
end_date	Ending date
unit	Specifies how the difference should be calculated

Important

Unlike DAYS, DATEDIF takes:

Start Date first → End Date second

This is another common interview trap.

Compare:

`=DAYS(end_date,start_date)`

versus:

`=DATEDIF(start_date,end_date,unit)`

Understanding DATEDIF Units

The most important units are:

Unit	Meaning
"Y"	Complete years
"M"	Complete months



"D"	Total days
"YM"	Remaining months after complete years
"MD"	Remaining days after complete months
"YD"	Remaining days after complete years

Let's understand them individually.

DATEDIF with "Y"

Purpose

Returns the number of **complete years** between two dates.

Syntax

=DATEDIF(start_date,end_date,"Y")

Business Example: Employee Tenure in Complete Years

In our dataset:

- I2 = Joining Date
- Current date = TODAY()

Formula:

=DATEDIF(I2,TODAY(),"Y")

Suppose:

Joining Date = 15-Mar-2021

Today = 10-Jul-2026

Result:

5

The employee has completed **5 full years**.

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DATEDIF with "M"

Purpose

Returns the total number of **complete months** between two dates.

Syntax

=DATEDIF(start_date,end_date,"M")

Example

=DATEDIF(I2,TODAY(),"M")

Possible result:

63

This means 63 complete months have passed.

Important

"M" returns **total complete months**, not the remaining months after years.



DATEDIF with "D"

Purpose

Returns the total number of days between two dates.

Syntax

```
=DATEDIF(start_date,end_date,"D")
```

Example Using the Dataset

```
=DATEDIF(D2,F2,"D")
```

This calculates total days from ticket start to completion.

For simple day differences, DAYS is generally clearer. DATEDIF becomes more valuable when calculating years, months, and combined durations.

DATEDIF with "YM"

Purpose

Returns the number of remaining months after complete years have been removed.

Suppose the duration is:

5 Years, 3 Months, 12 Days

Then:

```
=DATEDIF(start_date,end_date,"YM")
```

returns:

3

It ignores complete years.

DATEDIF with "MD"

Purpose

Returns the remaining number of days after complete months have been removed.

Suppose the duration is approximately:

5 Years, 3 Months, 12 Days

Then "MD" aims to return:

12

Important Caution

The "MD" unit can produce unintuitive results in certain end-of-month situations. For professional analytics, use it carefully and validate edge cases when exact month-end calculations matter.

DATEDIF with "YD"

Purpose

Returns the remaining number of days after complete years have been removed.

It ignores the year component and calculates the day difference within the remaining annual period.

Business Example: Employee Tenure in Years, Months and Days

Suppose the employee's Joining Date is in I2.

To calculate complete years:

=DATEDIF(I2,TODAY(),"Y")

To calculate remaining months:

=DATEDIF(I2,TODAY(),"YM")

To calculate remaining days:

=DATEDIF(I2,TODAY(),"MD")

Combine them:

=DATEDIF(I2,TODAY(),"Y")&" Years, "&DATEDIF(I2,TODAY(),"YM")&" Months, "&DATEDIF(I2,TODAY(),"MD")&" Days"

Possible result:

5 Years, 3 Months, 25 Days

This is one of the most common practical uses of DATEDIF.

Business Example: Employee Experience Category

Suppose:

- Less than 2 years → New Employee
- 2 to less than 5 years → Experienced
- 5 years or more → Senior Employee

Formula:

=IF(DATEDIF(I2,TODAY(),"Y")>=5,"Senior Employee",IF(DATEDIF(I2,TODAY(),"Y")>=2,"Experienced Employee","New Employee"))

This combines:

- DATEDIF
- TODAY
- Nested IF

It's a strong business scenario because it integrates functions from multiple lectures.

Business Example: Ticket Age

For an open ticket, suppose D2 contains the Start Date.

Formula:

=DATEDIF(D2,TODAY(),"D")

This returns how many days the ticket has remained open.

However, there is an important limitation: if the Start Date is in the future relative to TODAY(), DATEDIF returns an error because the start date cannot be later than the end date.

DAYS vs DATEDIF

Feature	DAYS	DATEDIF
Calculates days	Yes	Yes
Calculates complete months	No	Yes
Calculates complete years	No	Yes



Best for simple date differences	Yes	No
Best for employee tenure	No	Yes
Argument order	End, Start	Start, End
Unit argument required	No	Yes

Critical Interview Question: DAYS vs DATEDIF

Question

What is the difference between DAYS and DATEDIF?

Strong Answer

DAYS calculates the number of calendar days between two dates, while DATEDIF can calculate differences in complete years, months, days, or remaining date components. DAYS uses the Syntax DAYS(end_date,start_date), whereas DATEDIF uses DATEDIF(start_date,end_date,unit).

Module 5 Summary

Function	Purpose	Syntax
DAYS	Calculates calendar days between two dates	=DAYS(end_date,start_date)
DATEDIF	Calculates difference in years, months, or days	=DATEDIF(start_date,end_date,unit)

Core Formulas

=DAYS(F2,D2)

=DATEDIF(D2,F2,"D")

=DATEDIF(I2,TODAY(),"Y")

=DATEDIF(I2,TODAY(),"M")

=DATEDIF(I2,TODAY(),"Y")&" Years, "&DATEDIF(I2,TODAY(),"YM")&" Months"

Business Formulas

Ticket Status:

=IF(F2<=H2,"Completed On Time","Delayed")

Delay Days:

=MAX(0,DAYS(F2,H2))

Employee Tenure Category:

=IF(DATEDIF(I2,TODAY(),"Y")>=5,"Senior

Employee",IF(DATEDIF(I2,TODAY(),"Y")>=2,"Experienced Employee","New Employee"))

